

Software Setup Guide

Installing Pythia 8, HIJING, Geant4, and ROOT on your machine

Option A — Easiest: the sPHENIX Singularity container

If you have an SDCC (BNL) account or can pull the public container, this is the least painful path. Everything is pre-built.

```
# On RACF:
ssh <user>@sphnxlogin.sdcc.bnl.gov
source /cvmfs/sphenix.sdcc.bnl.gov/gcc-12.1.0/opt/sphenix/core/bin/sphenix_setup.sh -n new
source /cvmfs/sphenix.sdcc.bnl.gov/gcc-12.1.0/release/release_new/new.1/opt/sphenix/core/
bin/setup_local.sh \
/path/to/your/install
# Confirm it works:
which root
which Fun4All
pythia8-config --prefix
echo $G4INSTALL
```

Tip: You can also pull the publicly available singularity image and run it on any Linux box with singularity/apptainer installed. See sphenix-collaboration.github.io for the latest pull URL.

Option B — Docker container (any OS)

A pre-built image with Pythia8, HIJING, Geant4 and ROOT. Works on Windows, macOS, Linux.

```
# Install Docker Desktop first (docker.com/products/docker-desktop)
docker pull rootproject/root:6.30.04-ubuntu22.04
# Then run an interactive shell, mounting your work dir:
docker run -it --rm -v $PWD:/work -w /work \
rootproject/root:6.30.04-ubuntu22.04 bash
# Inside the container, install Pythia8 & Geant4 as in Option C below,
# OR use community images that already include them, e.g.:
docker pull hepstore/rivet:3.1.9 # Pythia8, Rivet, HepMC3 included
docker pull geant4/geant4:11.2.2 # Geant4 build image
```

Option C — Native install on Linux (Ubuntu 22.04+ / Debian)

C.1 System prerequisites

```
sudo apt update
sudo apt install -y build-essential cmake g++ gfortran git wget \
libxpm-dev libxft-dev libxext-dev libssl-dev \
libpcre3-dev libgsl-dev libxml2-dev libtbb-dev python3-dev \
libx11-dev libxmu-dev libexpat1-dev libxerces-c-dev \
qtbase5-dev libqt5opengl5-dev libqt5svg5-dev
```

C.2 ROOT

```
cd ~/src && git clone --branch v6-30-04 https://github.com/root-project/root.git root_src
mkdir -p ~/sw/root_build && cd ~/sw/root_build
cmake -DCMAKE_INSTALL_PREFIX=$HOME/sw/root ../../src/root_src
cmake --build . --target install -j$(nproc)
# Add to ~/.bashrc:
```

```
source $HOME/sw/root/bin/thisroot.sh
```

C.3 HepMC3 (event record used by Pythia/HIJING wrappers)

```
cd ~/src && git clone https://gitlab.cern.ch/hepmc/HepMC3.git
mkdir -p ~/sw/hepmc3_build && cd ~/sw/hepmc3_build
cmake -DCMAKE_INSTALL_PREFIX=$HOME/sw/hepmc3 \
-DHEPMC3_ENABLE_ROOTIO=ON -DHEPMC3_ENABLE_PYTHON=OFF \
.../src/HepMC3
make -j$(nproc) install
export HEPMC3_DIR=$HOME/sw/hepmc3
```

C.4 Pythia 8

```
cd ~/src && wget https://pythia.org/download/pythia83/pythia8310.tgz
tar xf pythia8310.tgz && cd pythia8310
./configure --prefix=$HOME/sw/pythia8 --with-hepmc3=$HOME/sw/hepmc3
make -j$(nproc) && make install
export PYTHIA8=$HOME/sw/pythia8
export PATH=$PYTHIA8/bin:$PATH
export LD_LIBRARY_PATH=$PYTHIA8/lib:$LD_LIBRARY_PATH
```

C.5 Geant4 11.2

```
cd ~/src && git clone --branch v11.2.2 https://github.com/Geant4/geant4.git
mkdir -p ~/sw/geant4_build && cd ~/sw/geant4_build
cmake -DCMAKE_INSTALL_PREFIX=$HOME/sw/geant4 \
-DGEANT4_INSTALL_DATA=ON \
-DGEANT4_USE_QT=ON -DGEANT4_USE_OPENGL_X11=ON \
-DGEANT4_USE_SYSTEM_EXPAT=ON \
.../src/geant4
make -j$(nproc) install
source $HOME/sw/geant4/bin/geant4.sh
```

C.6 HIJING (Fortran)

HIJING 1.383 is Fortran 77. It is distributed as source only from the original LBL site; the sPHENIX-patched copy is the most reliable starting point.

```
cd ~/src && git clone https://github.com/sPHENIX-Collaboration/HIJING.git
cd HIJING && make
# Produces ./hijing executable + libhijing.a
cp hijing $HOME/sw/bin/
```

Note: If you don't have the sPHENIX mirror, see the textbook reference in Module 5 for an alternative source tarball. You must have gfortran.

Option D — macOS (Homebrew)

```
# Install Xcode Command Line Tools first:
xcode-select --install
# Then Homebrew:
brew install cmake root gcc open-mpi xerces-c
# gcc is needed for gfortran (HIJING)
# Geant4 recipe:
brew install geant4
# Pythia8: build from source (Option C.4 works identically on macOS)
```

Tip: On Apple Silicon, Geant4's Qt visualization sometimes misbehaves. If so, build with `-DGEANT4_USE_QT=OFF` `-DGEANT4_USE_OPENGL_X11=OFF` and use only the

batch/command-line interface.

Option E — Windows via WSL2

Enable WSL2 and install Ubuntu 22.04 from the Microsoft Store, then follow Option C verbatim. For graphics (Geant4 visualization), use WSLg (Windows 11) or X410 / VcXsrv on Windows 10.

Environment file

Save this as `~/sphenix_course_env.sh` and source it at the start of every shell:

```
#!/bin/bash
export COURSE_ROOT=$HOME/sphenix_course
source $HOME/sw/root/bin/thisroot.sh
source $HOME/sw/geant4/bin/geant4.sh
export PYTHIA8=$HOME/sw/pythia8
export HEPMC3_DIR=$HOME/sw/hepmc3
export PATH=$PYTHIA8/bin:$HOME/sw/bin:$PATH
export LD_LIBRARY_PATH=$PYTHIA8/lib:$HEPMC3_DIR/lib:$LD_LIBRARY_PATH
echo "[env] Pythia8 @ $PYTHIA8"
echo "[env] Geant4 @ $G4INSTALL"
echo "[env] ROOT @ $ROOTSYS"
```

Verification checklist

- `root -l -q` opens a ROOT prompt and exits cleanly.
- `pythia8-config --version` prints 8.310 or later.
- `geant4-config --version` prints 11.2.x.
- hijing (or the patched binary) starts without segfault.
- `$HEPMC3_DIR/bin/HepMC3-config --version` prints 3.2.x.